

Smartphone Interface: Revised Prototypes

In Assignment I we worked divergently: we mapped the full problem space of a Bluetooth headphone companion app and produced low-fidelity paper prototypes that captured every function we could imagine an office user needing. In this stage we worked convergently. Rather than representing all the properties of the product, we narrowed the design down to the interactions that matter most to our persona and rebuilt them as high-fidelity, working prototypes (now under the product name Focus, continuing the HCI concept from Assignment I).

Every change below was evaluated against two questions carried over from our refined persona and interaction map: does this reduce the office professional's workload during a busy workday, and does this make the system's state and the available actions self-evident, so the user does not have to think? The revision is therefore framed throughout against Norman's design principles and Nielsen's ten usability heuristics, with the explicit goal of reducing user workload.

What Changed Between the Two Stages

The low-fidelity stage proved the information architecture; the high-fidelity stage refined it into an experience. The most significant changes are summarised below.

Area	Assignment I (low-fidelity)	Assignment II (high-fidelity)	Rationale (persona / usability)
Visual language	Hand-drawn paper wireframes; neutral, undefined styling	Restrained dark theme, generous spacing, single warm accent colour, real headphone render	Matches the persona's premium, discrete tool need; lowers cognitive load through visual calm
Main control screen	Sparse dashboard (battery, ANC, Live Translate, Meeting Mode)	Consolidated screen: battery %, Live Translate, Meeting Mode, ANC level, L/R balance, Max Volume Limiter, Auto Power Off	Surfaces the most frequent office controls at a glance; fewer taps to act
Secondary settings	Single Advanced Audio Settings panel	Progressive disclosure — EQ, device options and account settings revealed only on demand	Aesthetic and minimalist design; reduces visual noise
Account & support	Implicit / minimal	Dedicated Settings, Support (Quick Start Guide, FAQ, Help & Feedback) and Software Updates screens added	Help and documentation; recoverability
Destructive actions	Plain menu items	Confirmation dialogs added for Log Out, Forget Device, Power Off and Reset to Factory Settings	Error prevention; user control and freedom

Area	Assignment I (low-fidelity)	Assignment II (high-fidelity)	Rationale (persona / usability)
Connectivity feedback	Not represented	Explicit states: signal strength, You Are Offline message, multi-device connection status	Visibility of system status
Multi-device use	Not present	New multi-device connection screen with quick switching	Flexibility and efficiency for hybrid workers

In short, the revision removed ambiguity and added feedback rather than adding features. Several screens that were single, dense panels in the low-fidelity version were split so that the user only ever sees what the current task requires.

Decisions Driven by the Persona and the Interaction Map

Our refined persona is an office professional (roughly 20–55, moderate technical proficiency) working in a hybrid or open-plan environment, whose core need is to act on the most relevant control at a glance, without navigating deep menus during a busy workday. The revised interaction map keeps the hierarchy shallow — any primary action is reachable within roughly two taps — and groups content by frequency of use rather than by feature category.

This directly shaped two decisions. First, Live Translate, Meeting Mode and ANC remain on the primary control screen as one-tap controls, because they are the actions a meeting-bound user reaches for under time pressure. Second, lower-frequency tasks — equaliser tuning, firmware, device management, account settings — were pushed one level deeper behind clear entry points, so they never compete for attention with the primary controls. The interface presents exactly what is needed, when it is needed, which is the most concrete way we reduce the user's workload.

Applying Norman's Design Principles

- **Affordances and signifiers.** Every interactive element looks like what it does — toggles read as on/off switches, sliders (ANC, L/R balance) read as continuous adjustments, and the primary action on each screen is a filled accent button while secondary actions stay subdued.
- **Feedback.** Where the low-fidelity prototype was silent about state, the revision makes state continuously visible — battery percentage, signal strength, charging status and an explicit offline message. Every consequential tap produces an immediate, legible response.
- **Mapping.** The Left/Right balance slider is arranged so moving it left shifts audio left; the ANC control runs Low to High in the direction of increasing intensity. The relationship between control and outcome needs no explanation.
- **Constraints.** Error-prone paths are constrained: the verification field accepts only the expected input, password creation requires confirmation, and destructive actions are gated behind a deliberate confirmation step.

- **Conceptual model.** A consistent render of the headphone anchors the user's mental model — the app is clearly a controller for that physical object, making the on-screen controls feel like direct manipulation rather than abstract settings.

Applying Nielsen's Ten Usability Heuristics

We used Nielsen's heuristics as an explicit checklist when revising each screen.

1. **Visibility of system status.** Battery percentage, signal strength, charging state, the You Are Offline notice and live multi-device connection status keep the user continuously informed.
2. **Match between system and the real world.** Labels use the user's language — Meeting Mode, Live Translate, Auto Power Off — not engineering terms; Find Device uses a familiar map metaphor and the equaliser a recognisable frequency curve.
3. **User control and freedom.** Every screen carries a back control, and every destructive dialog (Log Out, Forget Device, Power Off) offers a clearly marked cancel as an emergency exit.
4. **Consistency and standards.** A single header pattern, a shared component set and consistent button styling run across all screens, following standard mobile platform conventions.
5. **Error prevention.** Irreversible actions are guarded by confirmation dialogs; the Max Volume Limiter prevents hearing-damaging levels and password confirmation prevents lock-out from typos.
6. **Recognition rather than recall.** Equaliser presets let users pick a recognisable option instead of recalling frequency values, and surfacing key modes on the home screen means users recognise their choices rather than recall where they are buried.
7. **Flexibility and efficiency of use.** Presets serve novices while Custom EQ serves power users; one-tap modes accelerate the common case; Auto Power Off offers quick presets; the multi-device screen lets hybrid workers switch sources quickly.
8. **Aesthetic and minimalist design.** The restrained palette, generous padding and progressive disclosure keep each screen focused on its primary task, so nothing competes with the information the user needs.
9. **Help users recognise, diagnose and recover from errors.** Edge states are explained in plain language with a path forward — the offline state states the cause and offers Go To Settings, and the verification screen offers a resend option.
10. **Help and documentation.** A dedicated Support section (Quick Start Guide, FAQ, Help & Feedback) plus Software Updates with release notes provide assistance without leaving the app.

Net Effect: Reduced User Workload

Taken together, the revisions reduce both physical and cognitive workload, which was the central instruction for this stage. Physical workload falls because the most frequent office

actions are reachable in fewer taps and the hierarchy stays shallow. Cognitive workload falls because the interface relies on recognition rather than recall, keeps system status continuously visible, maps controls naturally to their effects, and prevents errors before they require effortful recovery. Crucially, we achieved this by curating the experience around the persona rather than exposing every product capability — the interface communicates the device's most relevant behaviours clearly instead of mirroring its full feature list.

Smartwatch Interface: Companion Prototypes

Assignment I covered only the smartphone and tablet UI. In this stage we extended the system to a smartwatch companion. For our office-professional persona, the watch is the single most workload-reducing surface in the system: during a meeting or focused work the phone is often in a pocket or bag and the user's hands are on the keyboard. The watch lets them see headphone status and trigger the most time-critical actions — answer a call, mute, switch to Meeting Mode, start translation — from the wrist, without breaking focus. The watch was designed from the outset around the same refined persona and interaction map as the phone, but under far tighter constraints, which made Norman's and Nielsen's principles even more decisive.

Design Constraints That Shaped the Watch UI

The smartwatch screen is small, viewed in glances of one or two seconds, and operated on the move with a single finger or the digital crown. We therefore reduced each screen to a single primary action or a small grid of named shortcuts, used large circular touch targets, relied on colour and icon rather than text where possible, and grouped controls so the most office-critical actions sit at the top level. Where the phone can afford progressive disclosure across panels, the watch front-loads only what a busy user needs in the moment and pushes everything else behind a short scroll.

Refinement During This Stage

The watch interface was iterated within this stage. An early concept used a blue accent on lighter chrome with generic device labelling; we refined it toward a dark theme consistent with the Focus phone app, a shared accent system and — most importantly — colour-coded state feedback. This change does real usability work: status becomes legible at a glance without reading text, which is exactly what wrist use demands.

Aspect	Early concept	Refined version	Rationale
Theme	Blue accent on lighter chrome	Dark theme matching the Focus phone app	Cross-device consistency; lower glance effort
Mic state	Single mic screen with text label	Green when active, red when muted	Pre-attentive status; no reading required
Quick controls	Fixed grid	Scrollable grid: primary actions first, secondary on scroll	Keeps the primary glance uncluttered
Device labels	Generic (HQ-700 Pro, HQ Office PC)	Real targets (MacBook) with unified branding	Coherent product identity across devices

Decisions Driven by the Persona and the Interaction Map

The watch face doubles as a status surface: it shows the connected device and the currently active mode (for example, Connected · Meeting Mode) so the user confirms their state in a glance. The top-level quick grid mirrors the phone's home priorities — Meeting Mode, Mute, ANC and Translate — because these are the actions our persona reaches for under time

pressure; Bluetooth and Settings sit on a second scroll. Incoming calls surface a simple two-option screen (Answer / Reject) with large, colour-coded buttons, matching the urgency and one-handed nature of the moment.

Applying Norman's Design Principles

- **Affordances and signifiers.** Large circular buttons, clear toggles (Voice Clarity, Keyboard Noise Reduction) and radio lists (ANC Low/Mid/High, Noise Control) make each control's behaviour obvious.
- **Feedback.** The microphone screen turns green when active and red when muted; translation shows a live Listening waveform; the face shows battery and connection; Connection Lost and Reconnected states report system changes immediately.
- **Mapping.** The ANC list runs Low to High in order of increasing strength, and selection radios map directly to the chosen state.
- **Constraints.** Each screen offers only the few valid actions for its context, and radio lists prevent conflicting selections — important when a mis-tap is easy on a small screen.
- **Conceptual model.** The watch mirrors the headphone and the phone app, so the user carries one consistent mental model across all three devices.

Applying Nielsen's Ten Usability Heuristics

1. **Visibility of system status.** The face shows the connected device and active mode; status screens show battery, estimated playtime, mic active/muted colour and translation Listening state; connection changes are announced (Connection Lost, Reconnected).
2. **Match between system and the real world.** Plain labels (Meeting, Mute, Translate), the familiar TR to EN translation cue, and a microphone icon for capture state.
3. **User control and freedom.** Mute/Unmute, Pause translation, Answer/Reject and Reconnect give a clear action and exit at every step; standard watch gestures return to the previous screen.
4. **Consistency and standards.** The watch shares the phone app's dark theme and accent system and follows platform watch conventions (status bar with time, round-icon grid, crown scrolling).
5. **Error prevention.** The Low Battery Alert warns before the headphone dies; radio lists prevent invalid combinations; large targets reduce mis-taps on the small screen.
6. **Recognition rather than recall.** A labelled icon grid and the mode shown on the face let users recognise options instead of recalling buried settings.
7. **Flexibility and efficiency of use.** The headline gain: answering calls, muting, switching ANC and starting translation from the wrist removes the need to retrieve the phone at all; the scrollable grid keeps secondary actions one flick away.

8. **Aesthetic and minimalist design.** One primary action per screen, large legible type and a dark background keep each glance focused — a necessity, not a preference, at watch scale.
9. **Help users recognise, diagnose and recover from errors.** Device Connection Lost pairs the problem with a Reconnect action, and a green Reconnected / OK confirms recovery.
10. **Help and documentation.** Settings provides Quick Help and About directly on the wrist.

Net Effect: Reduced User Workload

The smartwatch is where the workload-reduction goal is most fully realised. It collapses multi-step phone interactions into a single glance or tap, surfaces status without any navigation, and uses colour-coded feedback so the user understands the system state pre-attentively. For an office professional mid-meeting, muting, switching to Meeting Mode or taking a call no longer interrupts their work — precisely the outcome Norman's and Nielsen's principles are meant to produce.

Working prototypes: the high-fidelity interactive prototypes for both the smartphone and smartwatch interfaces are available at the Figma link shared with this report.